

Product Requirements Document (PRD)

TapConnect

Version: 1.0

Product Type: Android networking application

Primary Value: Fast, private, context-aware real-world networking through NFC, BLE, and AI-powered insights

1. Product Summary

TapConnect is an Android networking app designed to make real-life introductions effortless. It allows users to connect instantly with people nearby using **NFC tap-to-connect** and **Bluetooth Low Energy (BLE) fallback discovery**. The app reduces the friction of exchanging social details, eliminates awkward manual searching, and adds AI-powered context so users know *why* they should connect.

TapConnect is built for high-density social and professional environments such as universities, hackathons, conferences, meetups, startup events, and creator communities.

2. Problem Statement

Real-world networking is often slow, awkward, and incomplete.

Key Pain Points

- People struggle to exchange profiles quickly in busy environments.
- Searching usernames or phone numbers is inefficient.
- QR code sharing is inconvenient and often breaks the flow of conversation.
- Traditional contact exchange lacks context, so users do not know why a connection matters.
- Networking tools usually focus on saving contact information, not improving the quality of the connection.
- Privacy is often weak because users must expose too much information too early.

Impact of the Problem

Because of these issues, people miss opportunities to connect, follow up, and collaborate. The experience feels slow and unnatural, especially in events where first impressions matter.

3. Product Vision

Make networking feel as natural as a tap, while making each connection more meaningful through privacy-first sharing and intelligent context.

4. Product Goals

1. Enable instant profile exchange in 1–2 interactions.
 2. Minimize networking friction at physical events.
 3. Support both NFC and BLE for reliable connectivity.
 4. Provide privacy controls for selective sharing.
 5. Offer AI-generated summaries and suggestions to improve connection quality.
 6. Make the product simple enough for first-time use in event environments.
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5. Target Users

Primary User Groups

- **Students:** university events, hackathons, clubs, campus networking
- **Professionals:** conferences, seminars, business events
- **Creators and freelancers:** collaborations, content partnerships, client discovery
- **Startup founders and builders:** investor meetups, accelerators, founder communities

User Needs

- Fast profile exchange
 - Better context about people nearby
 - Control over what gets shared
 - Easy discovery of relevant people
 - Low-effort follow-up after meeting someone
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6. Product Positioning

Unique Value Proposition

“The fastest and smartest way to connect with people around you.”

Differentiation

- NFC + BLE hybrid networking system
- AI-assisted networking insights
- Minimal interaction design
- Strong privacy-first sharing model

- Designed specifically for real-life networking events
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7. Solution Overview

TapConnect solves the networking problem through three layers:

7.1 Instant Proximity Connection

- NFC tap-to-connect for direct device exchange
- BLE fallback when NFC is unavailable
- Active networking mode to control discoverability

7.2 Intelligent Profile Understanding

- AI-generated bios and summaries
- Smart connection reasons
- Icebreaker suggestions
- Suggested priority matches nearby

7.3 Privacy and User Control

- User decides what profile data is shared
 - Only active users are discoverable
 - Connection requires intent
 - Undo feature for accidental connections
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8. Core Features and Requirements

8.1 Profile Creation and Management

Description

Users create a profile that represents them during networking sessions.

Profile Fields

- Full name
- Profile photo
- Short bio
- Role/title
- Organization or university
- Interests
- Social links:
- Instagram

- LinkedIn
- WhatsApp
- Optional future links such as GitHub, X, website

Requirements

- Users can edit their profile anytime.
- Users can choose which fields are public during networking.
- Default sharing preferences must be configurable.
- Profile data should be stored locally first, with optional backend support later.

Acceptance Criteria

- A user can create and save a profile successfully.
 - A user can preview what others will see before going live.
 - A user can update sharing preferences without recreating the profile.
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8.2 Networking Mode

Description

Networking Mode controls whether a user is visible to nearby users.

Behavior

- When ON, the user becomes discoverable through NFC/BLE.
- When OFF, the user is hidden from discovery.
- Users should clearly see current visibility status.

Requirements

- Simple ON/OFF toggle.
- Clear indicator when discoverable.
- Discoverability should only occur for active users.

Acceptance Criteria

- If Networking Mode is OFF, the user does not appear in nearby discovery.
 - If ON, the user becomes eligible for tap or nearby pairing.
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8.3 NFC Tap-to-Connect (Primary Feature)

Description

Two users can exchange identities instantly through a phone tap using NFC.

User Flow

1. User opens the app.
2. NFC is enabled.
3. User taps another compatible phone.
4. System exchanges a unique user ID or session token.
5. App fetches profile details.
6. A connection confirmation screen appears.
7. Shared profile is displayed.

UX Requirements

- Instant haptic feedback.
- Success animation on connection.
- Text feedback such as “Connected with [Name]”.
- Undo option for 3 seconds.
- Minimal steps and no manual search.

Technical Requirements

- NFC should be used for tap-based exchange.
- The app should exchange an identifier rather than expose full data directly.
- NFC flow should be secure, fast, and user-initiated.

Acceptance Criteria

- A tap triggers a successful exchange when both users are active.
 - The connection result is shown in under 2 seconds in normal conditions.
 - User can undo within the short undo window.
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8.4 BLE Nearby Discovery (Fallback Feature)

Description

BLE is used to detect nearby users when NFC is unavailable or unsupported.

User Flow

1. User opens the app.
2. App scans nearby BLE-enabled devices.
3. Nearby active users appear in a list.
4. User taps a profile.
5. Connection is created and shared profile is displayed.

UX Requirements

- Show nearby users with photo, name, and key context.

- Sort users by relevance or proximity.
- Indicate active/discoverable status.
- Make the list easy to scan in crowded environments.

Technical Requirements

- BLE scanning must respect Android permissions.
- Discovery should refresh automatically.
- Fallback must work when NFC is unavailable.

Acceptance Criteria

- Nearby active users appear when BLE is available.
- User can connect from the list with one tap.
- Scanning remains responsive without draining battery excessively.

8.5 Profile Sharing and Connection View

Description

After a connection, the app displays shared information in a clean and readable format.

Shared Data Options

- Name
- Photo
- Bio
- Role
- Interests
- Social links selected by the user

Requirements

- The profile display should be readable and modern.
- Shared information should respect privacy settings.
- Users should know exactly what they are sharing.

Acceptance Criteria

- Connection view shows only approved fields.
- Shared profile is easy to save or revisit later.

8.6 Privacy and Control

Description

Privacy is a core product principle.

Requirements

- No automatic sharing without user intent.
- Users must opt into Networking Mode.
- NFC tap and app open should confirm intent before sharing.
- Only active users are discoverable.
- Users control what appears on their networking profile.
- Users can disconnect or revoke visibility at any time.

Privacy Principles

- User-first consent
- Minimal exposure by default
- Temporary discoverability
- Selective sharing

Acceptance Criteria

- Users are not visible unless they enable Networking Mode.
 - No data is exposed beyond selected sharing preferences.
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8.7 AI Smart Networking

Description

AI enhances networking by adding context, clarity, and relevance.

AI Capabilities

1. Smart Profile Summary

Transforms raw profile details into a clean, professional summary.

2. Connection Suggestions

Identifies likely reasons two users should connect. Example: "You both are interested in AI and startups."

3. Icebreaker Suggestions

Provides short conversation starters based on profile overlap. Example: "Ask about their recent project."

4. Smart Prioritization

Ranks nearby users by relevance instead of showing only raw proximity.

Requirements

- AI content should be concise and useful.
- AI should enhance rather than overwhelm the experience.
- Users should still be able to use the app without AI.
- AI features can be optional in early versions and enabled progressively.

Acceptance Criteria

- App can generate a readable summary from profile data.
 - Nearby users can be ranked based on shared interests or relevance.
 - Icebreaker suggestions appear quickly and clearly.
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9. User Personas

Persona 1: University Student

Needs to exchange details quickly after meeting classmates, mentors, or event speakers.

Persona 2: Conference Attendee

Wants fast professional networking without typing usernames or scanning QR codes.

Persona 3: Freelancer / Creator

Wants to share portfolio and social handles with potential collaborators.

Persona 4: Founder / Builder

Needs fast and meaningful connections with relevant people at events.

10. User Experience and Main Screens

10.1 Profile Setup Screen

Purpose: create or edit profile.

Contains:

- Name

- Photo upload
- Bio
- Interests
- Social links
- Sharing preferences
- Save button

10.2 Networking Mode Screen

Purpose: control visibility and discoverability.

Contains:

- On/off toggle
- Status indicator
- NFC/BLE availability indicator
- Short privacy explanation

10.3 Nearby Users Screen

Purpose: show discoverable users nearby.

Contains:

- User list
- Photo and name
- Interest match indicator
- Sort/filter option in future versions
- Connect button

10.4 Connection Animation Screen

Purpose: confirm successful connection.

Contains:

- Tap or connect animation
- Success message
- Undo button
- Short profile preview

10.5 Profile View Screen

Purpose: display received profile details.

Contains:

- Name and photo
 - Bio
 - Shared links
 - AI summary
 - Icebreaker suggestion
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11. UX Principles

- Minimal clicks: target 1–2 interactions
 - Fast response: under 2 seconds for core actions where possible
 - Clean modern interface
 - Strong visual feedback: animation, vibration, status changes
 - Low cognitive load: no unnecessary menus
 - Mobile-first and event-friendly design
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12. Technical Architecture

12.1 Platform

- Android
- Kotlin

12.2 Connectivity Layer

NFC

- Used for direct tap-based exchange
- Primary instant connection mechanism

BLE

- Used for nearby device discovery
- Fallback when NFC is not available

12.3 Data Handling

- Local storage for user profile and preferences
- Optional backend for future scale and multi-device sync
- Use unique user IDs or session tokens for exchange

12.4 Data Flow

1. User activates Networking Mode.
2. Device becomes discoverable.
3. Another user taps via NFC or discovers via BLE.
4. User ID/session token is exchanged.
5. App fetches profile data.
6. Shared profile is rendered.
7. Connection is stored locally or in backend later.

12.5 Security Considerations

- Encrypt sensitive stored data where possible.
 - Exchange identifiers instead of raw private data.
 - Use consent-driven visibility.
 - Limit discovery to active sessions.
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13. MVP Scope

Must-Have for First Version

- User profile creation
- Profile editing
- Networking Mode toggle
- BLE nearby discovery
- Tap-to-connect experience for BLE or basic nearby flow
- Shared profile display UI
- Privacy controls
- Local data storage
- Connection confirmation and undo

Optional / Phase 2

- NFC support
- AI profile summaries
- AI icebreakers
- AI ranking/prioritization
- Backend sync
- Cross-device history

MVP Goal

Deliver a usable networking experience with minimal friction and strong privacy, while keeping AI and advanced backend features for later expansion.

14. Future Enhancements

- Event-based networking rooms
 - AI matchmaking based on shared goals and interests
 - Cross-platform support for iOS
 - In-app chat after connection
 - Connection analytics and event insights
 - Calendar and follow-up reminders
 - Export to CRM or contact tools
 - Smart badge or organizer integrations
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15. Success Metrics

Product Metrics

- Time to connect: under 2 seconds
- Successful connection rate
- Number of connections per event
- Number of active discoverable users
- Retention rate after event use

Engagement Metrics

- Profile completion rate
- Number of profile views
- Number of accepted connections
- Percentage of users using AI suggestions

Quality Metrics

- Connection success without errors
 - BLE discovery reliability
 - App crash rate
 - Permission acceptance rate
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16. Risks and Challenges

Technical Risks

- BLE complexity across Android devices
- NFC compatibility differences by device
- Android permission handling
- Battery consumption during discovery
- Background scan limitations

Product Risks

- Users may not enable Networking Mode
- AI suggestions may feel generic if not well tuned
- Too much UI complexity may reduce speed
- Privacy misunderstandings could reduce trust

Mitigation Strategies

- Keep onboarding simple
 - Show clear permission prompts and explanations
 - Prioritize one-tap actions
 - Introduce AI gradually
 - Test on multiple Android devices
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17. Assumptions

- Users will use TapConnect in event environments.
 - Android will be the first supported platform.
 - Users are willing to create a lightweight networking profile.
 - NFC and BLE access will be allowed on supported devices.
 - A later backend can be introduced if scale requires it.
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18. Out of Scope for MVP

- Full social feed
 - Public content posting
 - Advanced messaging system
 - Multi-platform release
 - Large-scale analytics dashboard
 - Complex matching algorithm requiring server-side AI infrastructure
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19. Release Strategy

Phase 1: MVP

Focus on profile creation, networking mode, BLE discovery, and connection display.

Phase 2: Proximity Expansion

Add NFC, stronger reliability, and improved connect flow.

Phase 3: AI Enhancement

Add summaries, icebreakers, prioritization, and smarter suggestions.

Phase 4: Growth Features

Add event rooms, chat, analytics, and multi-platform support.

20. Final Product Statement

TapConnect is a next-generation networking app that removes friction from real-world introductions, uses proximity technologies like NFC and BLE, and improves every connection with privacy-first design and AI-powered context.

It is built to make networking faster, smarter, and more natural.